# module6_lab2

October 11, 2026

[1]:
```
import sqlite3
```

[2]:
```
import pandas as pd
```

[3]:
```
con = sqlite3.connect("world.db")
```

[4]:
```
#creating a table 2.2 - adding drop table per professor in slack
```

[5]:
```
con.execute("DROP TABLE IF EXISTS City;")
```

[5]:
```
<sqlite3.Cursor at 0x1b69287f7c0>
```

[6]:
```
con.execute("""
CREATE TABLE 'City' (
'ID' INTEGER PRIMARY KEY AUTOINCREMENT,
'Name' char(35) NOT NULL default '',
'CountryCode' char(3) NOT NULL default'',
'District' char(20) NOT NULL default'',
'Population' int(11) NOT NULL default '0'
);""")
```

[6]:
```
<sqlite3.Cursor at 0x1b68ffc5740>
```

[7]:
```
df_city = pd.read_sql("SELECT * FROM City", con)
print(df_city)
```

```
Empty DataFrame
Columns: [ID, Name, CountryCode, District, Population]
Index: []
```

[8]:
```
# insert at least 8 fictional rows 2.3
```

[9]:
```
con.execute("INSERT INTO City VALUES (NULL, 'New York', 'USA', 'New York',␣
 ↪10000000);")
con.execute("INSERT INTO City VALUES (NULL, 'Los Angeles', 'USA', 'California',␣
 ↪4000000);")
con.execute("INSERT INTO City VALUES (NULL, 'Orlando', 'USA', 'Florida',␣
 ↪2700000);")
```

```
con.execute("INSERT INTO City VALUES (NULL, 'Dallas', 'USA', 'Texas', 2300000);
 ↪")
con.execute("INSERT INTO City VALUES (NULL, 'Las Vegas', 'USA', 'Nevada',␣
 ↪1600000);")
con.execute("INSERT INTO City VALUES (NULL, 'Columbus', 'USA', 'Ohio', 1500000);
 ↪")
con.execute("INSERT INTO City VALUES (NULL, 'Atlanta', 'USA', 'Georgia',␣
 ↪1400000);")
con.execute("INSERT INTO City VALUES (NULL, 'Pittsburgh', 'USA',␣
 ↪'Pennslyvania', 1400000);")
con.execute("INSERT INTO City VALUES (NULL, 'Salt Lake City', 'USA', 'Utah',␣
 ↪1200000);")
con.commit()
```

[10]:
```
df_city = pd.read_sql("SELECT * FROM City", con)
display(df_city)
```

| | ID | Name | CountryCode | District | Population |
|---|---|---|---|---|---|
| 0 | 1 | New York | USA | New York | 10000000 |
| 1 | 2 | Los Angeles | USA | California | 4000000 |
| 2 | 3 | Orlando | USA | Florida | 2700000 |
| 3 | 4 | Dallas | USA | Texas | 2300000 |
| 4 | 5 | Las Vegas | USA | Nevada | 1600000 |
| 5 | 6 | Columbus | USA | Ohio | 1500000 |
| 6 | 7 | Atlanta | USA | Georgia | 1400000 |
| 7 | 8 | Pittsburgh | USA | Pennslyvania | 1400000 |
| 8 | 9 | Salt Lake City | USA | Utah | 1200000 |

[11]:
```
#markdown cell - One markdown note on a MySQL-to-SQLite difference you handled
# In MySQL, it willl accept an empty string such as "", but in SQlite, you␣
 ↪actually have to tpye out NULL or you'll run into an error
```